Machine Learning

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For any square matrix A, a vector v is called an eigen vector iff

$$Av = \lambda v$$

where λ is a scaler and
λ is eigen value of the given eigen vector v

- Eigen vectors capture the variance in data along their directions
- the amount of variance captured by them is proportional to their eigen values

$$x_1 \; x_2 \; x_3 \ldots x_{20} \longrightarrow \boxed{\text{PCA}} \longrightarrow P_1 \; P_2 \; P_3 \ldots P_{20}$$

$$\text{data matrix} = X = \begin{bmatrix} x_{11} & x_{21} & \cdots & x_{n1} \\ x_{12} & x_{22} & & x_{n2} \\ x_{13} & x_{23} & & x_{n3} \\ x_{14} & x_{24} & & x_{n4} \\ \vdots & \vdots & & \vdots \\ x_{120} & x_{220} & & x_{n20} \end{bmatrix}_{20 \times n}$$

create covariance matrix of X

$$\text{cov}(X) = \underset{20 \times n}{X} \; \underset{n \times 20}{X^T} = S \leftarrow \text{dim} = 20 \times 20$$

$\hookrightarrow$ square matrix

covariance matrix helps in finding eigen vectors which maximize the variance

$$SV = \lambda V$$

we'll find all eigen vectors and their respective eigen values using this formula

for 20×20 covariance matrix, we'll get 20 eigen vectors

e_1	$\lambda_1 = 2$	⋮	
e_2	$\lambda_2 = 4$	⋮	
e_3	$\lambda_3 = 0.5$	⋮	
e_4	$\lambda_4 = 50$	e_{19}	$\lambda_{19} = 1.5$
e_5	$\lambda_5 = 5$	e_{20}	$\lambda_{20} = 2.1$
⋮			

sort eigen vectors in descending order of eigen values

$e_4 > e_{10} > e_5 > e_2 \ldots$

now original data has dim = 20 or no. of features = 20

lets say we decided to reduce these dimensions to 5 for this we'll select top 5 eigen vectors and transform the original data using them

$$W = \begin{bmatrix} e_4 \\ e_{10} \\ e_5 \\ e_2 \\ e_{20} \end{bmatrix} \longrightarrow \text{matrix of top 5 eigen vectors}$$

$\leftarrow$ dim = 5 × 20

$$W = \begin{bmatrix} e_4 \\ e_{10} \\ e_5 \\ e_7 \\ e_{20} \end{bmatrix}$$

5×20

5 eigen vectors as rows

each eigen vector has 20 values

$$\underset{5\times n}{Z} = \underset{5\times 20}{W}\ \underset{20\times n}{X}$$

X - original data matrix

W - eigen vector matrix

Z - transformed data matrix

$$\underset{\text{20 features}}{X} \longrightarrow \overset{W}{\boxed{PCA}} \longrightarrow \underset{\text{5 features}}{Z}$$

$X = [d_1 \quad d_2 \quad d_3 \ldots d_n]$ each datapoint as a column vector

$$W = \begin{bmatrix} e_4 \\ e_{10} \\ e_5 \\ e_7 \\ e_{20} \end{bmatrix}$$

each eigen vector as a row vector

$$wx = \begin{bmatrix} e_4 \\ e_{10} \\ e_5 \\ e_2 \\ e_{20} \end{bmatrix} \begin{bmatrix} d_1 & d_2 & d_3 & \dots & d_n \end{bmatrix}$$

$$Z = \begin{bmatrix} e_4 . d_1 & e_4 . d_2 & e_4 . d_3 & \dots & e_4 . d_n \\ e_{10} . d_1 & e_{10} . d_2 & e_{10} . d_3 & \dots & e_{10} . d_n \\ \vdots & & & & \\ e_{20} . d_1 & e_{20} . d_2 & \dots & & e_{20} . d_n \end{bmatrix}_{5 \times n}$$

this is how using PCA we reduced dimensions of data from 20 to 5

Real use case -

$$x_1 \; x_2 \dots x_{20} \longrightarrow \boxed{\text{log regr}} \longrightarrow \hat{y}$$

due to too many features, this model was overfiting

$$x_1 \; x_2 \dots x_{20} \longrightarrow \boxed{\text{PCA}} \longrightarrow \underbrace{p_1 \; p_2 \; p_3 \; p_4 \; p_5}$$

5 principle components using top 5 eigen vectors

$$p_1 \; p_2 \; p_3 \; p_4 \; p_5 \longrightarrow \boxed{\text{log regr}} \longrightarrow \hat{y}$$

we'll check if this reduced overfiting or not

Hyperparameter Tuning using cross validation:

Linear Regression

- learning α needs HP tuning

$$w_{new} = w_{old} - \alpha \frac{\partial L}{\partial w_{old}}$$

$\alpha = 0.1, 0.01, 0.02, 10^{-3}, 2 \times 10^{-3} \ldots$

we need to choose optimal value of α using HP tuning

K fold Cross Validation:

lets say $K = 4$ for Kfold KV

this will divide train data into 4 subsets

train data $\longrightarrow$ f_1 f_2 f_3 f_4

each containing 25% of the data

for $\alpha = 0.1$ and

data folds – f_1 f_2 f_3 f_4

train 4 different lin regr models

model	train data	validation data	validation matrix (r^2)
m_1	f_2 f_3 f_4	f_1	r_1
m_2	f_1 f_3 f_4	f_2	r_2
m_3	f_1 f_2 f_4	f_3	r_3
m_4	f_1 f_2 f_3	f_4	r_4

final r2 score for $\alpha = 0.1$ = avg(r_1 r_2 r_3 r_4) = $r_{\alpha=0.1}$

similarly we'll train 4 models for every α and compute their average r2 scores on validation sets

5 values of α and $K=4$ cross validation

total models trained = $5 \times 4 = 20$

α	avg r2 score
0.1	$r_{\alpha=0.1}$
0.01	$r_{\alpha=0.01}$
0.02	$r_{\alpha=0.02}$
0.04	$r_{\alpha=0.04}$
0.08	$r_{\alpha=0.08}$

choose the α which gives best avg r2 score

lets say $\alpha = 0.04$ gives the best avg r2 score then we set $\alpha = 0.04$

Grid Search Cross Validation:

eg. DT model

HP	values to test
max depth	[3, 4, 5, 6]
min samples split	[5, 7, 9]
min samples leaf	[4, 6]

HP setup :

lets say $k = 5$

max depth	min samples split	min samples leaf	no. of models	avg validation metric
3	5	4	5	a_1
3	5	6	5	a_2
3	7	4	5	a_3
3	7	6	5	a_4
3	9	4	5	a_5
3	9	6	5	⋮
4	5	4	5	⋮
4	5	6	⋮	⋮
4	7	4	⋮	⋮
4	7	6	⋮	⋮
4	9	4	⋮	⋮
4	9	6		
⋮	⋮	⋮		

the HP combination which gives the best average validation metric, will be selected as optimal hyperparameters.